INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 6

1. Experiment Title:

Sorting Rows and Removing Repeats

2. Aim:

Sort product rows, display unique categories, and show the two most expensive products.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

ORDER BY sorts query results. DESC sorts from highest to lowest. DISTINCT removes duplicate values from the result. LIMIT restricts the number of returned rows in supported SQL systems.

5. Algorithm / Procedure:

1. Create the product table and insert the four products.
2. Sort product names and prices in descending price order.
3. Select distinct categories and sort them alphabetically.
4. Select the two highest-priced products using ORDER BY and LIMIT.

6. Program / Queries / Commands:

```
CREATE TABLE product (
    product_name VARCHAR(30),
    category VARCHAR(20),
    price INT
);

INSERT INTO product (product_name, category, price)
VALUES
    ('Pen', 'Stationery', 15),
    ('Notebook', 'Stationery', 60),
    ('Calculator', 'Electronics', 850),
    ('Bag', 'Office', 500);

-- Task 1
SELECT product_name, price FROM product
ORDER BY price DESC;

-- Task 2
SELECT DISTINCT category FROM product
ORDER BY category;

-- Task 3
SELECT product_name, price FROM product
ORDER BY price DESC
LIMIT 2;
```

7. Input:

Pen—15, Notebook—60, Calculator—850, Bag—500.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: product_name	price
Calculator	850
Bag	500
Notebook	60
Pen	15
Task 2: category	
Electronics	
Office	
Stationery	
Task 3: product_name	price
Calculator	850
Bag	500

9. Result:

The SQL statements for Experiment 6 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is ORDER BY used for?
2. What is the difference between ASC and DESC?
3. What does DISTINCT do?
4. What does LIMIT 2 mean?